

Blind validation of an automated colony counter

What held up, what didn't, and why

Manual colony counting is slow, and it gives you a number with no context. A tech counts a plate, writes down 47, and that's it: no flag on the odd satellite colony near the edge, no record of which colonies were touching, no repeatability across shifts. Existing automated counters fix the speed problem but mostly stop there. You get a count, not an assessment.

This year I've been building an instrument that does more: automated bacterial colony counting with morphological anomaly detection and an organism-aware profile system, in Dr. Eric Ryberg's lab as a CAPS Research Scholar at UConn. This is a validation memo: I ran the system blind against synthetic data it had never seen, and I'm reporting the real numbers, including the ones that missed our target.

AT A GLANCE

1.000 Precision on every plate. Zero false colonies.	0.869 Pooled F1 across 250 blind plates (target 0.90)	3 of 5 Held-out sets at or above target	15.75% Mean count error (5.31% on passing sets)
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System architecture

The full stack, built from the register level up:

- **Hardware.** Custom power electronics on a KiCad-designed PCB, driving plate illumination and imaging.
- **Vision.** An OpenCV computer-vision pipeline on a Raspberry Pi 5, running from illumination correction through segmentation to morphometric analysis.
- **Anomaly detection.** Two layers: geometric flagging (area, circularity, and aspect-ratio outliers) plus a statistical/ML layer.
- **Organism profiles.** Counting and flagging thresholds adapt per organism instead of applying one fixed set of assumptions to every plate.

Blind validation methodology

I generated 250 synthetic plates across five held-out parameter sets (density, illumination, size variance, morphology, and touching rate), 50 plates per set, using combinations never seen during pipeline development. Ground truth (exact colony positions, radii, and anomaly labels) was embedded in the generator output.

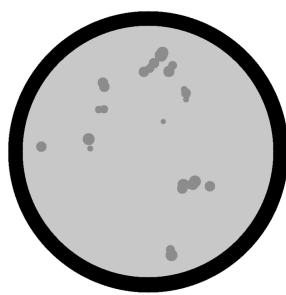
Critically: `quantify_colonies()` was called with only the image path. Ground truth was loaded strictly afterward, only to score what the pipeline had already returned. It never informed detection. That ordering is what makes this validation blind.

Matching used one-to-one nearest-centroid assignment with a gate at $1.5\times$ the ground-truth colony's radius, not raw count comparison. Otherwise a false positive and a missed colony can cancel out and look like a perfect score. Results are pooled (micro-averaged) across all plates in a set rather than averaged per plate, so a handful of near-empty plates can't swing the aggregate.

I benchmarked against peer-reviewed numbers for commercial and open-source counters, not vendor claims: OpenCFU at 50.31% mean count error in an independent re-benchmark (Chen et al. 2025), and ColonyDoc-It at 18.3–59.7% depending on agar medium (Heuser et al. 2023).

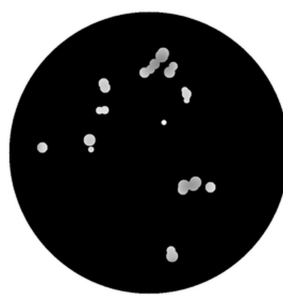
The CV pipeline

Figure 1 follows one plate through every stage. I deliberately used the hardest plate in the run, `high_touching/plate_039` ($F1 = 0.513$), so the walkthrough shows the pipeline under stress rather than at its best. The same plate comes back in the results.



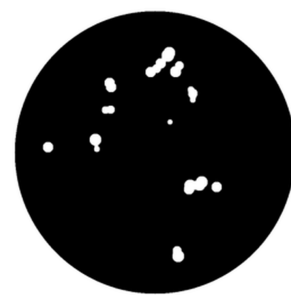
1 Raw input

A held-out synthetic plate, straight from the generator. The pipeline receives only this image.



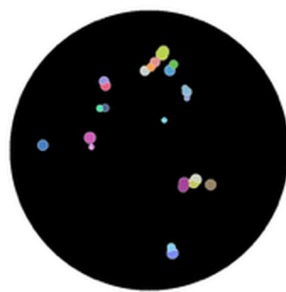
2 Illumination correction

Normalizes uneven lighting across the dish before anything downstream depends on absolute pixel intensity.



3 Threshold

Isolates colonies from background. Every later stage depends on clean separation here.



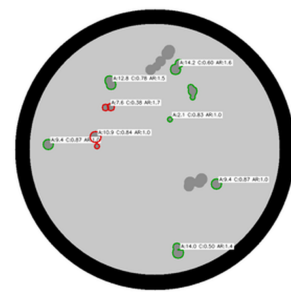
4 Watershed segmentation

Splits touching colonies, seeded from local maxima in the distance transform. This stage decides whether two overlapping colonies count as two or one.



5 Morphometrics

Extracts area, circularity, and aspect ratio for each colony from its segmented contour.



6 Anomaly detection

Flags outliers (red): colonies whose shape or size falls outside the expected range for the organism profile.

Figure 1. The pipeline, stage by stage, on the lowest-scoring plate in the run. Grey clusters with no outline (panels 5–6) were not counted. Watershed does produce regions inside them (panel 4), but a downstream filter removes them. Which filter, and whether those regions are real colonies or fragments, is the next diagnostic.

Results

Overall pooled F1 across all 250 plates was **0.869**, against a target of 0.90. Three of the five held-out sets passed. Two did not.



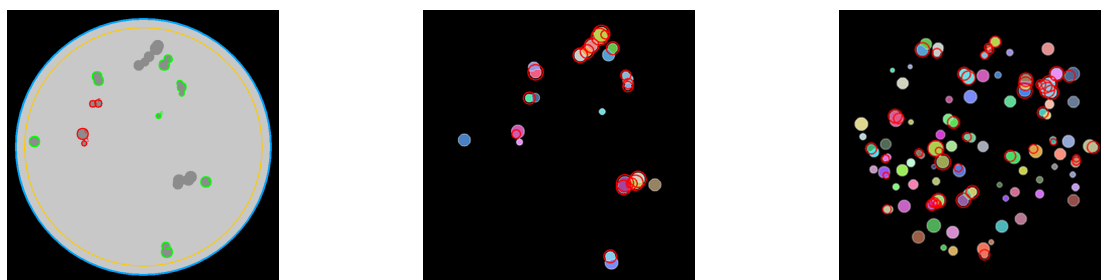
Figure 2. Pooled F1 by held-out set (50 plates each); dashed line marks the 0.90 target. Both failing sets stress colonies that touch.

One number matters more than any of those. **Precision was 1.000 on every set and every plate.** Across all 250 plates, the pipeline never fabricated a colony. Every point of F1 lost is a real colony missed, never a false one counted.

Mean count error was 15.75% pooled across all five sets and 5.31% on the three passing sets. Both are below the published error rates above (18.3–59.7%). Those benchmarks were measured on real agar and mine on synthetic plates, so this is context, not a head-to-head comparison. A direct comparison requires running both systems on identical plates under identical conditions, which is what biological validation is for.

The worst plate

The single worst plate in the run scored F1 = 0.513 with recall of 0.345. Precision stayed at 1.000. Figure 3 puts what the pipeline reported next to what it missed.



a What it reported

10 colonies counted (green), 3 flagged as anomalies (red). Every one of them is real.

b What it missed

Missed colonies circled in red. Nearly all sit inside touching clusters.

c Same failure, denser plate

dense_pack/plate_021. The same failure at a higher colony count, so it isn't specific to one plate.

Figure 3. (a–b) high_touching/plate_039, the same plate as Figure 1. The whole F1 loss is recall. (c) A plate from the other failing set.

Why two sets fail

When colonies overlap heavily enough, the merged blob's distance transform has only one local maximum. There's no second peak for watershed, or any marker-seeding technique, to find, regardless of parameter tuning. I confirmed this directly by sampling distance-transform values between touching colony centers rather than inferring it from the F1 drop.

All three pipeline parameters in scope for this iteration (`min_area_mm2`, watershed marker seeding, and `bg_blur_kernel`) were swept exhaustively against this failure mode with no meaningful movement. Closing the gap needs a different segmentation approach, such as Hough-circle fitting or concave-point contour splitting inside merged blobs. That's new architecture, not a parameter change.

For perspective: the FDA BAM countable range (25–250 CFU/plate) has an upper limit because crowded plates are hard to count, for humans too. Yet `dense_pack`'s 80–120 colonies per plate (Figure 3c) sit well inside that “countable” range and still break segmentation. It's a known industry-wide hard case, not a naive implementation gap. It's still a gap, and I'm reporting it, not excusing it.

Bugs found and fixed along the way

Two meaningful bugs were found and fixed during this run. I'm listing them because reproducible research means documenting what changed, not just what the final system does.

- 1. Per-blob watershed threshold.** Watershed's foreground threshold was computed against the whole plate's peak distance value instead of per blob, silently dropping markers on mixed-size plates. Fixing it moved F1 from 0.804 to 0.872 (measured before the area fix below).
- 2. A regression from that fix.** The fix undersized isolated colonies' measured area by about 10%. A separate area-accuracy CI check caught it, and it was fixed before the branch merged.

Anomaly detection: deferred

Anomaly detection is weak across the board (pooled F1 of 0.234), and I did not tune it this cycle. Colony counting hadn't converged on two of the five sets, so I deferred it rather than layering it on an unstable base. The red flags in Figures 1 and 3 show the mechanism working, not that it's accurate yet.

What's next

The quickest win. 99.2% of `size_variance`'s missed colonies exceed a fixed `max_area_mm2` filter. Deriving that limit from plate calibration, not a hardcoded constant, should close most of that gap fast.

Biological validation. The next major phase moves to real clinical plates instead of synthetic renders. Everything above is synthetic by design, so the pipeline still has to prove itself against real agar, real bubbles, and real staining before any of these numbers mean anything clinically. That's the next memo.

Contributors welcome. The full pipeline is open source: github.com/hmbirmingham/imaging-system. If you work with colony morphology and want to contribute an organism profile (expected size, circularity, and anomaly thresholds for a specific organism), I'd like the help. That's the part built to be extended, and it's not something I can populate alone.